Non-linear Diction: Linear Representations Do Not Predict Linear Steerability for Emotional Styles

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Abstract

Steering vectors—directions in activation space that causally shift model behavior—are an increasingly popular tool for controllable generation and AI safety. They rest on the assumption that concepts are represented as linear directions, a view formalized by the Linear Representation Hypothesis. But does linear *detectability* imply linear *steerability*? We investigate this question for 11 emotional styles in PYTHIA-2.8B by comparing three measures: (i) linear versus MLP probe accuracy, (ii) contrastive activation addition effectiveness, and (iii) prompting success. We find that all 11 styles are represented linearly (nonlinearity indices 0.965–1.008), yet linear separability does not predict steering effectiveness ($r = -0.220$, $p = 0.52$). Prompting succeeds uniformly for all styles (3.8–4.4/5 adherence), including those that steer poorly, and is uncorrelated with steering success ($r = -0.018$, $p = 0.96$). Our results reveal a *detection–control gap*: styles that are linearly separable in activation space are not necessarily linearly controllable, suggesting that the model's read-out and generation pathways involve nonlinear computation that prompting can access but activation addition cannot.

1 Introduction

Steering vectors have become a standard tool for controlling language model behavior at inference time [Turner et al., 2023, Rimsky et al., 2024]. The core idea is simple: identify a direction in activation space that corresponds to a concept, then add a scaled version of that direction during generation to shift output toward the target concept. This approach rests on a strong assumption—that concepts are encoded as linear directions in the model's residual stream. The Linear Representation Hypothesis (LRH) formalizes this view, and substantial evidence supports it for many concept types [Park et al., 2023, Jiang et al., 2024].

Yet a persistent puzzle remains. Some styles and behaviors resist linear steering despite appearing to be "known" by the model. Konen et al. [2024] find that disgust achieves near-random steering accuracy (AUC 0.51), and surprise yields few successful steering vectors, even though language models can readily produce disgusted or surprised text when prompted. If the model can generate a style on demand via prompting, why would a linear intervention in activation space fail? One possibility is that these styles are encoded *non-linearly*, making them invisible to linear probes and steering vectors alike. Another is that the representation is linear but the *causal pathway* from representation to generation is not.

We disentangle these possibilities by conducting the first systematic comparison of three facets of style encoding across a broad taxonomy of 11 emotional styles: (i) *linear separability*, measured by comparing linear and MLP probes; (ii) *steerability*, measured by contrastive activation addition (CAA); and (iii) *promptability*, measured by style adherence ratings. We test these on PYTHIA-2.8B using the GOEMOTIONS dataset [Demszky et al., 2020].

Our central finding is a *detection–control gap*: all 11 styles are linearly separable in activation space (non-linearity indices ≈ 1.0), yet steering effectiveness varies dramatically and is uncorrelated with

linear separability ($r = -0.220$, $p = 0.52$). Meanwhile, prompting succeeds uniformly for all styles (100% success rate, 3.8–4.4/5 adherence), including those that steer poorly. The correlation between steering and prompting success is essentially zero ($r = -0.018$, $p = 0.96$).

These results have direct implications for AI safety and controllable generation. Linear intervention methods—steering vectors, sparse autoencoder features—may systematically miss behaviors that are linearly detectable but causally accessed through nonlinear pathways. Our contributions are:

- We show that all 11 tested emotional styles are linearly represented in PYTHIA-2.8B (MLP probes never outperform linear probes), supporting the LRH for representation geometry.
- We demonstrate that linear separability does not predict steering effectiveness, revealing a detection–control gap where representation geometry and causal intervention diverge.
- We establish that prompting and steering access fundamentally different pathways to style control, with prompting succeeding uniformly where steering fails selectively.

The rest of this paper is organized as follows. Section 2 reviews related work on steering vectors, the linear representation hypothesis, and non-linear representations. Section 3 describes our experimental methodology. Section 4 presents our main results. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Work

The Linear Representation Hypothesis. The idea that concepts correspond to directions in activation space has been formalized by Park et al. [2023], who unify three interpretations—subspace, measurement, and intervention—under a causal inner product framework. Testing 27 concepts on LLaMA-2-7B, they find linear structure for 26, with the sole exception being the relational concept `thing→part`. Jiang et al. [2024] provide a theoretical account: linearity arises from the interaction of next-token prediction with softmax and cross-entropy, but this analysis assumes binary concepts with clear contrasts. Park et al. [2024] extend the LRH to categorical concepts (polytopes) and hierarchical concepts (orthogonal subspaces), showing that even within the linear framework, complex concept types require richer geometric structures. Our work tests whether the LRH holds for emotional styles and, crucially, whether linear detectability implies linear controllability.

Steering vectors and activation engineering. Turner et al. [2023] introduce Activation Addition (ACTADD), computing steering vectors from contrastive prompt pairs and adding them to the residual stream. They demonstrate effective steering for sentiment and topics but observe failures for compositional concepts and non-monotonic behavior when using partial dimensions. Rimsky et al. [2024] scale this approach via Contrastive Activation Addition (CAA), averaging activation differences across many examples and testing on behavioral dimensions (sycophancy, corrigibility) in Llama 2 Chat. Konen et al. [2024] apply both activation-based and training-based style vectors to emotions and writing styles in Alpaca-7B, finding that disgust and surprise resist steering while joy and anger succeed. Unlike these works, we systematically compare steering effectiveness against linear separability and prompting success across the same style taxonomy.

Non-linear representations. Csordás et al. [2024] discover "onion representations" in small GRUs, where tokens at different positions share the same direction but differ in magnitude, requiring nonlinear decoding. Linear probing and distributed alignment search completely fail on these representations while nonlinear interventions succeed. Elhage et al. [2022] show that neural networks store more features than dimensions via superposition, organizing them into geometric polytope structures. This implies that features sharing dimensions cannot be cleanly separated by linear methods. Whether similar non-linear encodings exist for style in large transformers remains unexplored—our probing experiments directly test this.

Benchmarking steering methods. Wu et al. [2025] introduce AxBench, the first large-scale benchmark comparing steering methods across 500 concepts in Gemma-2-2B/9B. Their key finding is that prompting outperforms all representation-based steering methods, including sparse autoencoders and diff-in-means vectors. This "prompting dominance" is consistent with our results but does not explain *why* prompting succeeds where linear methods fail. Our work provides a mechanistic perspective: the detection–control gap suggests that prompting accesses style through computational pathways that bypass the limitations of single-direction activation addition. Han et al. [2023] find that style control through output embedding transformations requires a full $d \times d$ matrix rather than

a single direction, further suggesting that style involves higher-dimensional structure than a single linear direction can capture.

Style transfer and controlled generation. The broader literature on style transfer and controlled generation provides context for our style taxonomy. Demszky et al. [2020] introduce GoEmotions, the dataset we use, with 54K Reddit comments annotated for 27 emotion categories. Chalnev et al. [2024] use sparse autoencoders to measure causal effects of steering vectors and find that naive linear steering produces unpredictable side effects, consistent with our observation that the relationship between activation-space metrics and behavioral effects is nonlinear.

3 Methodology

We conduct three parallel experiments measuring different facets of style representation in PYTHIA-2.8B: probing (does the model represent each style linearly?), steering (can a linear vector causally shift output?), and prompting (can the model produce each style via instruction?). We then correlate all three measurements across 11 emotional styles.

3.1 Model and Data

Model. We use Pythia-2.8B [Biderman et al., 2023], a 32-layer transformer with $d_{\text{model}} = 2560$, accessed via TransformerLens [Nanda and Bloom, 2022]. Pythia is non-gated, well-supported by mechanistic interpretability tools, and large enough to encode style while remaining computationally tractable.

Dataset. We use GOEMOTIONS [Demszky et al., 2020], a corpus of 54K Reddit comments annotated for 27 emotion categories plus neutral. We filter to single-label examples (excluding multi-label instances for clean signal), retain up to 500 examples per category, and require a minimum of 150 examples for inclusion. This yields 11 styles plus neutral, with sample sizes ranging from 488 (fear) to 500 (most categories). Texts are truncated to 128 tokens and processed with padding and attention masks.

Styles tested. We select 11 styles spanning a range of expected steering difficulty based on prior work: joy, anger, sadness, fear, disgust, surprise, love, gratitude, admiration, curiosity, and confusion. Disgust and surprise are expected to be hard to steer based on Konen et al. [2024]; joy and anger are expected to be easy; the remainder are exploratory.

3.2 Experiment 1: Probing

For each style s and layer $\ell \in \{4, 8, 12, 16, 20, 24, 28, 31\}$, we:

1. Collect mean-pooled residual stream activations $\boldsymbol{h}_\ell(x) = \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{h}_\ell^{(t)}(x)$ for texts labeled with style s and for neutral texts.
2. Balance classes to equal size.
3. Train a logistic regression probe (linear, $C = 1.0$) and a two-layer MLP probe (hidden sizes 256 and 128, with early stopping on 15% validation data) using 5-fold cross-validation.
4. Record accuracy and compute the non-linearity index: $\text{NL}(s) = \text{acc}_{\text{MLP}}(s)/\text{acc}_{\text{linear}}(s)$.

A non-linearity index significantly greater than 1.0 indicates non-linear encoding; a value near 1.0 indicates that linear probes capture all available information.

3.3 Experiment 2: Steering

For each style s, we extract a steering vector at the layer with the highest linear probe accuracy:

$$\boldsymbol{v}_s = \frac{1}{|\mathcal{D}_s|}\sum_{x\in\mathcal{D}_s} \boldsymbol{h}_\ell(x) - \frac{1}{|\mathcal{D}_{\text{neutral}}|}\sum_{x\in\mathcal{D}_{\text{neutral}}} \boldsymbol{h}_\ell(x) \tag{1}$$

We evaluate steering vectors along two dimensions:

Table 1: Probing results at the best layer per style. Linear and MLP probe accuracies are reported with the non-linearity index (NL Index = MLP/Linear). NL Index ≈ 1.0 for all styles indicates uniformly linear representations. Best linear probe accuracy in **bold**.

Style	Layer	Linear Acc.	MLP Acc.	NL Index
Gratitude	16	**0.970**	0.954	0.984
Love	4	0.939	0.934	0.995
Fear	4	0.900	0.881	0.980
Surprise	8	0.892	0.895	1.003
Joy	4	0.872	0.855	0.981
Admiration	8	0.872	0.874	1.002
Sadness	4	0.856	0.826	0.965
Disgust	4	0.847	0.827	0.976
Curiosity	20	0.831	0.838	1.008
Anger	12	0.826	0.828	1.002
Confusion	12	0.807	0.809	1.002

Activation-space metrics. We compute Cohen's d (effect size of the mean difference), steering vector classification accuracy (using $\boldsymbol{v}_s$ as a linear classifier), and cosine separation between style and neutral activation centroids.

Generation metrics. We generate text from 15 diverse prompts with and without the steering vector added to the residual stream at scaling factor $\alpha = 3.0$ and temperature 0.7. We classify each generated text using a logistic regression probe trained on the original activations and measure the shift in style classification rate relative to unsteered baseline.

Random vector control. We steer with random vectors of the same norm as each real steering vector to confirm specificity. We report the difference in style projection between real and random vectors.

3.4 Experiment 3: Prompting

For each style s, we prompt GPT-4.1-nano with "Write in a [style] tone. Continue: [prompt]" for 10 diverse prompts. We use GPT-4.1-nano as a judge to rate style adherence on a 1–5 scale and check specificity by rating against a random alternative style. We also generate baseline text with no style instruction for comparison.

Model mismatch. Probing and steering use PYTHIA-2.8B while prompting uses GPT-4.1-nano, because Pythia-2.8B is a base model with poor instruction-following. We discuss this limitation in section 5.4.

3.5 Correlation Analysis

We compute Pearson correlations between all pairs of: linear probe accuracy, MLP probe accuracy, non-linearity index, Cohen's d, steering vector accuracy, generation shift, prompting score, and prompting specificity. We report p-values and assess significance at $\alpha = 0.05$.

4 Results

4.1 All Styles Are Linearly Represented

Table 1 shows probing results at each style's best layer. Non-linearity indices range from 0.965 (sadness) to 1.008 (curiosity), with a mean of 0.991. The MLP probe does not outperform the linear probe for any style at any layer. This result is unambiguous: PYTHIA-2.8B represents all 11 tested emotions as linear directions in activation space.

Despite uniform linearity, linear separability varies substantially across styles. Gratitude (0.970) and love (0.939) are the most separable, while confusion (0.807) and anger (0.826) are the least. This 0.16-point range reflects genuine differences in how distinctly each style is encoded.

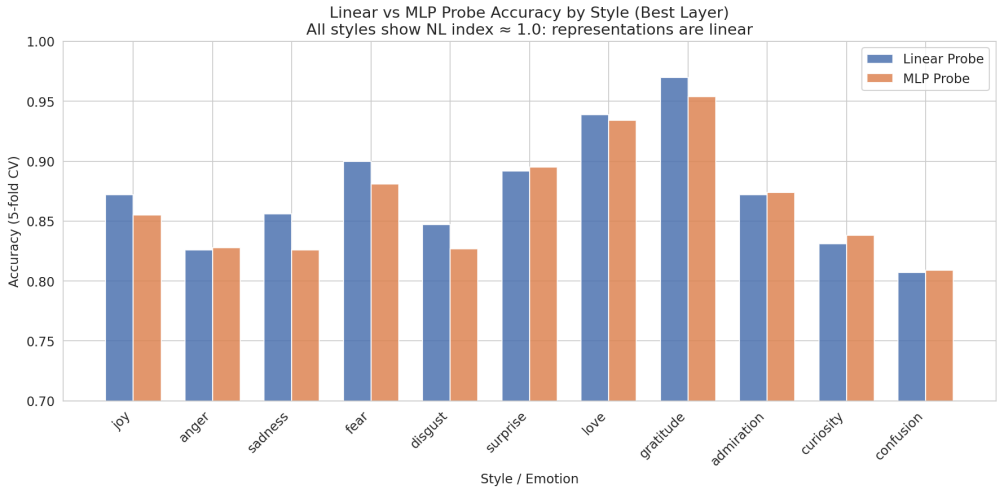


Figure 1: Linear versus MLP probe accuracy for each style. The two probe types achieve nearly identical accuracy across all styles, with non-linearity indices clustered tightly around 1.0. This indicates that MLP probes extract no additional information beyond what linear probes capture.

Table 2: Steering results for each style. Cohen's d measures activation-space separation; generation shift measures the change in style classification rate relative to unsteered baseline. Largest generation shifts in **bold**. Linear separability does not predict either metric.

Style	Cohen's d	SV Acc.	Baseline	Steered	Shift
Gratitude	0.381	0.569	0.067	0.800	**+0.733**
Fear	0.090	0.499	0.400	0.800	+0.400
Love	0.147	0.512	0.200	0.533	+0.333
Joy	0.634	0.620	0.333	0.600	+0.267
Anger	0.129	0.504	0.133	0.400	+0.267
Confusion	0.455	0.587	0.333	0.600	+0.267
Disgust	0.129	0.510	0.400	0.533	+0.133
Curiosity	0.525	0.604	0.200	0.333	+0.133
Sadness	0.907	0.673	0.333	0.267	−0.067
Surprise	0.163	0.539	0.467	0.400	−0.067
Admiration	0.192	0.534	0.600	0.533	−0.067

Layer-wise patterns. Most styles achieve peak probe accuracy at early layers (layer 4), but gratitude peaks at layer 16 and curiosity at layer 20, suggesting that some emotional dimensions are computed in deeper layers. Figure 1 shows the comparison of linear and MLP probes across styles.

4.2 Steering Effectiveness Is Independent of Linear Separability

Table 2 presents steering results. Cohen's d ranges from 0.090 (fear) to 0.907 (sadness), and generation shift ranges from -0.067 to $+0.733$. The correlation between linear probe accuracy and Cohen's d is $r = -0.220$ ($p = 0.52$)—weak and non-significant.

Several results are noteworthy. Sadness has the highest Cohen's d (0.907) yet a *negative* generation shift (-0.067): the mean-difference vector points in a direction that is linearly separable but not causally effective for generation. Fear has the lowest Cohen's d (0.090) yet the second-largest generation shift (+0.400), suggesting that even weak steering vectors can cause large behavioral shifts when they align with sensitive decision boundaries. Gratitude achieves the largest generation shift (+0.733) despite only moderate activation-space separation ($d = 0.381$).

Random vector control. Random vectors of the same norm produce negligible style projection changes (-0.023 for joy, -0.241 for anger, -0.028 for sadness), confirming that real steering vectors carry style-specific information (figure 2).

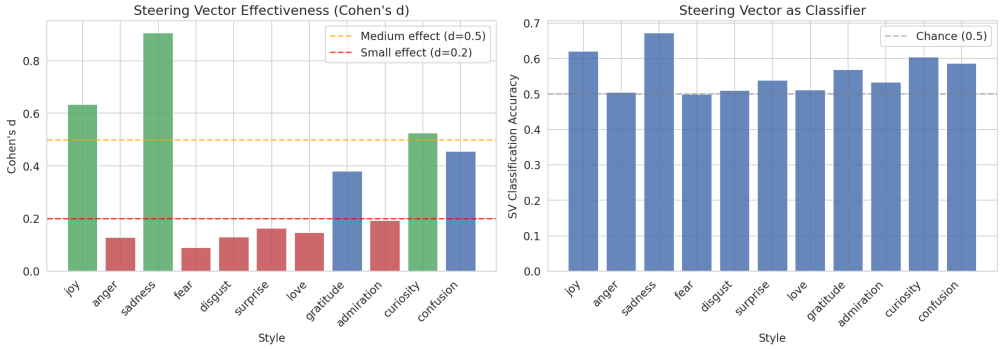


Figure 2: Steering effectiveness measured by Cohen's d and steering vector classification accuracy per style. Styles with high activation-space metrics (sadness) do not necessarily produce positive generation shifts, while styles with low metrics (fear) sometimes do.

Table 3: Prompting results using GPT-4.1-nano. Target score is the mean style adherence rating (1–5). Specificity is the difference between target style rating and random alternative style rating. All styles achieve 100% success rate. Best target scores in **bold**.

Style	Target Score	Success Rate	Specificity
Joy	**4.40**	100%	+3.30
Disgust	**4.40**	100%	+0.70
Gratitude	**4.40**	100%	+2.20
Surprise	4.10	100%	+1.90
Anger	4.00	100%	+2.40
Sadness	4.00	100%	+1.80
Fear	4.00	100%	+1.50
Love	4.00	100%	+3.00
Admiration	4.00	100%	+3.00
Curiosity	3.90	100%	+0.70
Confusion	3.80	100%	+1.80

4.3 Prompting Succeeds Uniformly

Table 3 shows prompting results. All 11 styles achieve 100% success rate ($\geq 3/5$ adherence score) with mean adherence ranging from 3.8 (confusion) to 4.4 (joy, disgust, gratitude). Specificity scores confirm that prompted text is rated higher for the target style than for a random alternative style.

Notably, disgust—historically the hardest style to steer [Konen et al., 2024]—achieves the joint-highest prompting score (4.4/5). This directly demonstrates that steering difficulty does not predict prompting difficulty.

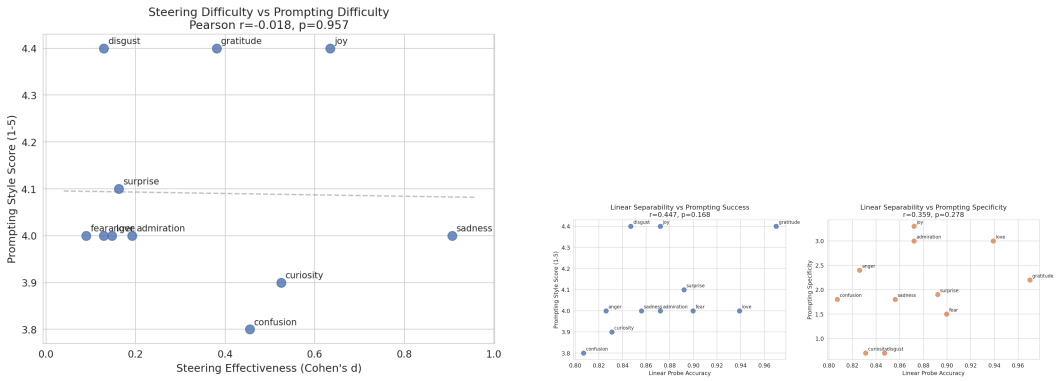
4.4 The Three Measures Are Independent

Table 4 presents the full correlation matrix across our three measurement dimensions. No correlation reaches statistical significance.

The most striking result is the near-zero correlation between steering effectiveness and prompting success ($r = -0.018$, $p = 0.96$). Figure 3 visualizes this relationship. The three measures—linear separability, steerability, and promptability—are largely independent dimensions of how the model encodes and accesses style information.

Table 4: Pearson correlations between probing, steering, and prompting metrics. None reach statistical significance ($\alpha = 0.05$), indicating that the three measures capture largely independent dimensions of style encoding.

Correlation	Pearson r	p-value
Linear acc. $\leftrightarrow$ Prompt score	0.447	0.168
Linear acc. $\leftrightarrow$ Prompt specificity	0.359	0.278
NL index $\leftrightarrow$ Prompt score	-0.515	0.105
Cohen's $d \leftrightarrow$ Linear acc.	-0.220	0.516
Cohen's $d \leftrightarrow$ Prompt score	-0.018	0.957
SV acc. $\leftrightarrow$ Prompt score	-0.029	0.932



(a) Steering vs. prompting

(b) Linearity vs. prompting

Figure 3: *(Left)*Steering effectiveness (Cohen's d) versus prompting adherence score for each style. The near-zero correlation ($r = -0.018$) shows these access fundamentally different pathways. *(Right)*Linear probe accuracy versus prompting score; the weak positive trend ($r = 0.447$) is not significant.

5 Discussion

5.1 The Detection–Control Gap

Our central finding is a dissociation between detecting a style in activation space and controlling it via activation addition. We call this the *detection–control gap*. A style can be linearly separable—meaning a probe can reliably distinguish it from neutral—without being linearly steerable, meaning that adding the mean-difference vector to the residual stream does not reliably shift generation toward that style.

This gap is most vivid for sadness, which has the highest Cohen's d (0.907) among all styles yet produces a *negative* generation shift (-0.067). The mean-difference vector correctly identifies the direction that separates sadness from neutral activations, but adding this direction during generation does not cause the model to produce sadder text. The representation is "read-only linear"—detectable but not causally effective through simple additive intervention.

Why does the gap exist? We hypothesize that the model's generation process involves nonlinear computation between activation directions and output token probabilities. Even if a style is encoded as a linear direction, the model may read this direction through a sequence of nonlinear transformations (attention, MLP layers, layer normalization) that collectively implement a complex function. Activation addition modifies the direction but does not replicate the full computational pathway that maps style representations to output distributions. Prompting, by contrast, sets up the entire forward pass to produce styled text, engaging all intermediate computations.

This interpretation is consistent with the finding of Wu et al. [2025] that prompting outperforms all representation-based steering methods across 500 concepts. It also aligns with Chalnev et al. [2024],

who show that naive steering produces unpredictable side effects—a symptom of the gap between the linear direction being modified and the nonlinear function that reads it.

5.2 Implications for the Linear Representation Hypothesis

Our results support the LRH as a statement about representation *geometry*: all 11 emotional styles are linearly separable, with non-linearity indices tightly clustered around 1.0. However, the LRH is sometimes interpreted more broadly as implying that linear interventions should be effective for any linearly-represented concept. Our results challenge this stronger interpretation. Linear separability is necessary but not sufficient for linear steerability. The LRH should be understood as a statement about how information is encoded, not about how it is causally used.

This distinction echoes Park et al. [2023], who formalize separate "measurement" and "intervention" interpretations of the LRH. Our probing results support the measurement interpretation (linear probes detect style), while our steering results show that the intervention interpretation does not follow automatically.

5.3 Implications for AI Safety

Steering vectors and sparse autoencoder features are increasingly used for AI safety applications, including reducing sycophancy, hallucination, and harmful behaviors [Rimsky et al., 2024]. Our detection–control gap suggests a systematic limitation of these approaches: behaviors that are linearly detectable may not be linearly controllable. If this gap extends to safety-relevant behavioral dimensions—which we have not tested—then linear intervention methods may provide incomplete coverage even when linear probes suggest the behavior is well-represented.

Prompting remains the most reliable method for style control, consistent with AxBench findings [Wu et al., 2025]. However, prompting cannot be applied in all safety-relevant contexts (e.g., when the user controls the prompt). Developing non-linear steering methods that close the detection–control gap is an important direction for safety research.

5.4 Limitations

Model mismatch. Probing and steering use PYTHIA-2.8B while prompting uses GPT-4.1-nano. This is because Pythia-2.8B is a base model with poor instruction-following ability. Ideally, all three experiments would use the same model. Our prompting results demonstrate that the styles are promptable *in general*, but the specific prompting–steering comparison is cross-model. Testing on instruction-tuned models where both steering and prompting can be applied would strengthen this comparison.

Small generation sample. We use only 15 prompts for steering evaluation, introducing substantial variance. Several styles show generation shifts within the noise floor (± 0.067). Larger sample sizes would provide more reliable effect estimates.

Emotion as style proxy. GoEmotions labels are crowd-sourced emotion annotations on Reddit comments, which are noisy proxies for "style" as understood in literary or linguistic terms. True stylistic dimensions (irony, formality, stream-of-consciousness) would be a stronger test but lack standard labeled datasets at sufficient scale.

Single model. Our results are specific to Pythia-2.8B. Larger models, instruction-tuned models, or architecturally different models (e.g., with gated MLPs) may exhibit different patterns. The detection–control gap could be smaller or larger depending on model capacity and training.

Binary probing. We test style-versus-neutral rather than multi-class probing across all styles simultaneously. Multi-class probing might reveal interference patterns or non-linear structure that binary probing misses.

6 Conclusion

We investigated whether the model represents emotional styles linearly and whether linear separability predicts steerability and promptability. Across 11 emotional styles in PYTHIA-2.8B, we

find that all are linearly represented (non-linearity indices 0.965–1.008) but that linear separability does not predict steering effectiveness ($r = -0.220$, $p = 0.52$). Prompting succeeds uniformly for all styles (3.8–4.4/5), including those that steer poorly, and is uncorrelated with steering success ($r = -0.018$, $p = 0.96$). These results reveal a detection–control gap: style representations are linear, but the causal pathway from representation to generation is not.

Our work has two practical takeaways. First, for practitioners seeking controllable generation, prompting remains the most reliable and universal method for style control. Second, for researchers developing linear intervention methods for AI safety, linear probe accuracy should not be taken as evidence that a behavior is linearly controllable—the detection–control gap means that some behaviors may resist linear steering even when they are linearly detectable.

Future work should test whether the detection–control gap extends to safety-relevant behavioral dimensions (sycophancy, deception, harmful content) and whether non-linear steering methods (e.g., trained steering vectors, distributed alignment search) can close the gap. Testing on instruction-tuned models where both steering and prompting operate on the same model would eliminate the model mismatch limitation. Expanding the style taxonomy beyond emotions to literary and compositional styles would further probe the boundaries of the LRH.

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